

Roll No.: _____

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First Semester Int.M.Sc. Maths MDS / S1 Int. M.Sc. Physics SC

24CSA108 - Hands on Introduction to Computing I

Mid-Term Examination, October 2025

Time: Two hours

Maximum marks: 50

Answer all questions

Qn		Marks	CO	BTL
1	Predict the output of the following code snippets. If there is an error, correct the code and then write the output. (a) <code>for i in range(2, 8, 2):</code> <code>print (i * '*')</code> (b) <code>def greet(name):</code> <code>return "Hello, " + name</code> <code>print(greet("World"))</code> (c) <code>x = input("Enter a number")</code> <code>if x % 2 = 0:</code> <code>print("Even number")</code> <code>else</code> <code>print("Odd number")</code> (d) <code>print("Hello" + 5)</code> (e) <code>total = 0</code> <code>for i in range(1, 6):</code> <code>if i % 2 != 0:</code> <code>total += i</code> <code>print(total)</code> (f) <code>def check_age(age):</code> <code>if age >= 18:</code> <code>return "Adult"</code> <code>return "Minor"</code> <code>print(check_age(15))</code> (g) <code>for i in range(1, 10):</code> <code>if i % 2 == 0:</code> <code>continue</code> <code>print(i)</code> (h) <code>total = 0</code> <code>for i in range(1, 5)</code> <code>total = total + i</code> <code>print("Sum is: " total)</code> (i) <code>x = 5</code> <code>y = 10</code> <code>z = x > y or x + 5 == y</code> <code>print(z)</code> (j) <code>for i in range(1, 3):</code> <code>for j in range(2):</code> <code>print(i+j)</code>	2 × 10	CO2	L2

2	Evaluate the following expressions and give an explanation for the result obtained (a) $-2 ** 3 ** 2$ (b) $2 - 5\% - 3$ (c) $-7//4$ (d) $10 // 3 + 3 ** 2 - 4$ (e) <code>type (complex(2,3).real is int</code>	2×5	CO1	L1
3	Write a python program to swap the values of two variables a and b without using a third variable.	5	CO1	L3
4	Write a python program that takes two numbers as input, performs addition, subtraction, multiplication, and division (handle division by zero with a message), and prints results using if-else for zero check.	5	CO3	L1
5	Implement a program based on flowchart logic: Take a number, use a for loop to sum its digits (e.g., $123 \rightarrow 1 + 2 + 3 = 6$), and if the sum is even, print "Even sum", else "Odd sum".	5	CO3	L3
6	Write a program with a function <code>print_pattern(rows)</code> that takes an integer 'rows' and uses nested for loops to print a triangle pattern of numbers. The output for <code>print_pattern(3)</code> is <pre>1 1 2 1 2 3</pre>	5	CO3	L2

L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create